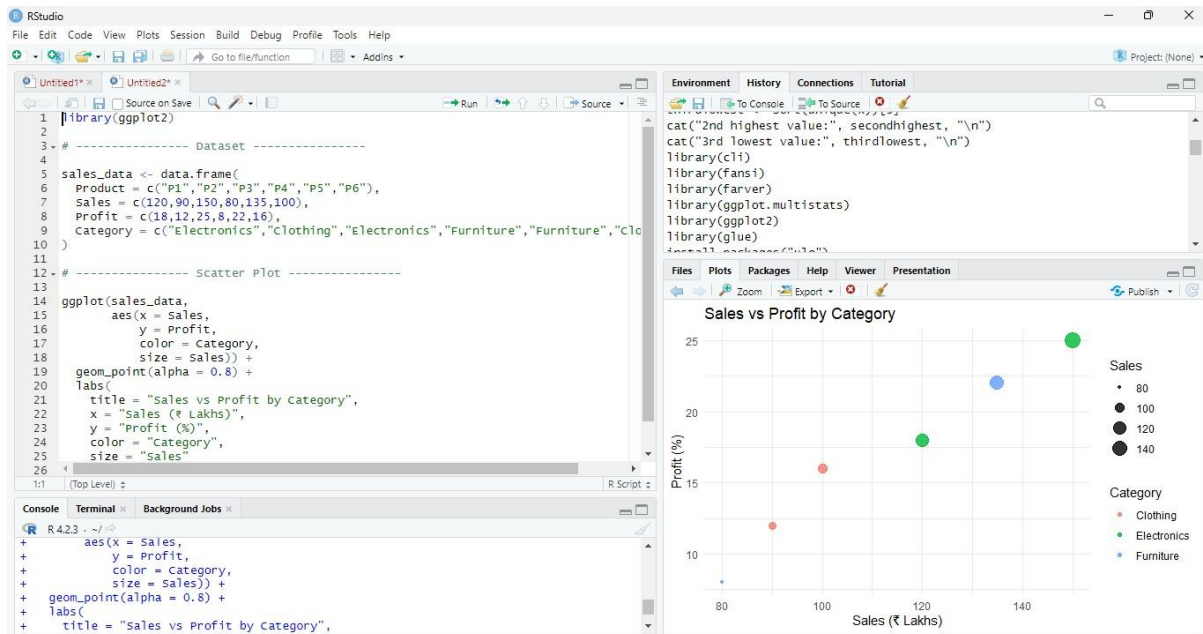


Data Interpretation Exercise

Q1)

1. Write an R program to create a scatter plot with Sales on the x-axis and Profit on the y-axis.



3. Interpretation of the relationship between Sales and Profit

- The scatter plot shows a **positive relationship** between Sales and Profit.
- As **Sales increase, Profit also tends to increase.**
- Electronics products have the highest sales and profit.
- Furniture and Clothing products have comparatively lower values.

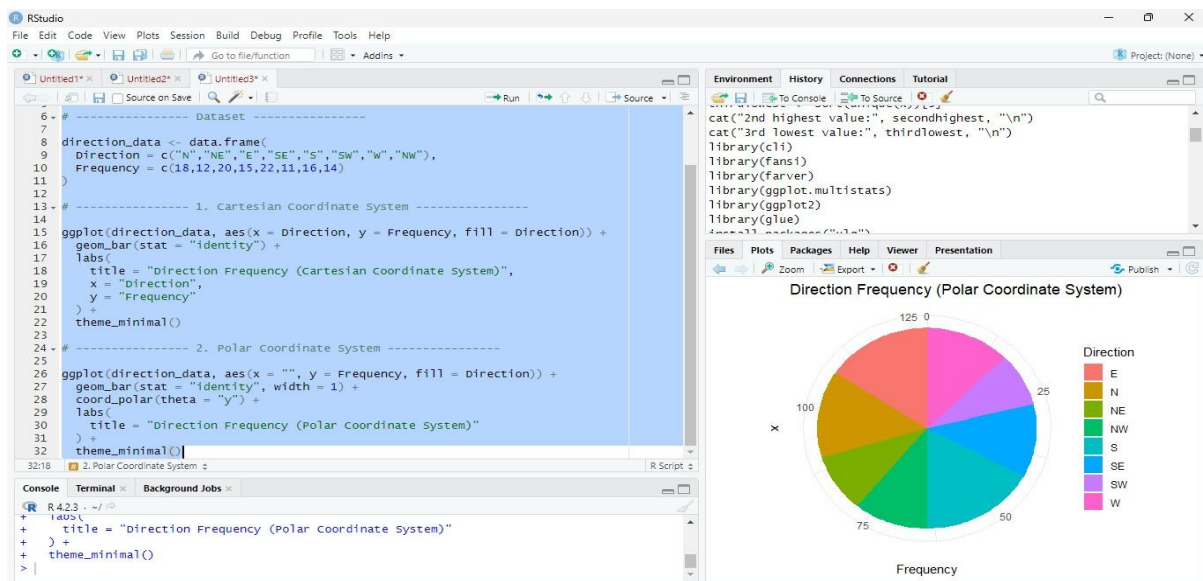
4. Type of Scale Used

Variable	Scale Type
Sales	Continuous (Numeric)
Profit	Continuous (Numeric)
Category	Categorical (Nominal)
Color	Mapped to Category
Size	Mapped to Sales

5. Alternative Visualization

- A **bubble chart** (already created by mapping `size = Sales`) is a good alternative because it shows Sales, Profit, and Category together.
- Another suitable alternative is a **grouped bar chart** to compare Sales and Profit across categories.

Q2) Coordinate Systems (Cartesian and Polar)



3. Comparison of Both Visualizations

- **Cartesian Coordinate System (Bar Chart):**
 - Easy to compare frequencies.
 - Differences between directions are clearly visible.
 - Best for accurate value comparison.
- **Polar Coordinate System (Pie Chart):**
 - Displays the proportion of each direction.
 - More visually appealing for directional or circular data.
 - Less accurate for comparing similar values.

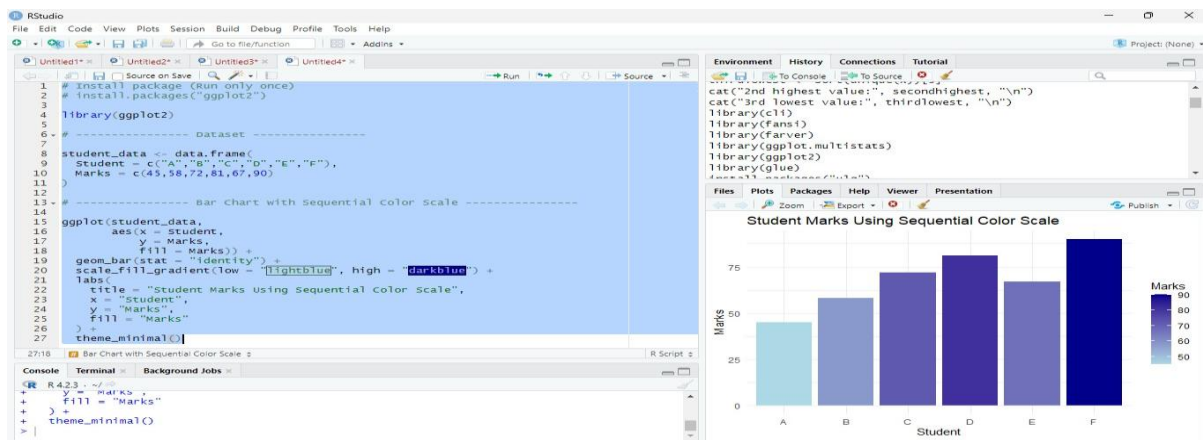
4. Which Coordinate System is More Appropriate?

Polar Coordinate System is more appropriate for directional data.

5. Justification

- Directional data (North, South, East, West, etc.) naturally represents **angles around a circle**.
- The polar coordinate system reflects compass directions, making it easier to understand the distribution of frequencies.
- The Cartesian coordinate system is better for comparing exact values, while the polar coordinate system better represents the circular nature of directional data.

Q3) Color Scales for Data Representation



2. Highest and Lowest Marks

- **Highest Marks:** Student F → 90
- **Lowest Marks:** Student A → 45

3. Why is a Sequential Color Scale Appropriate?

- Marks are **continuous numeric values**.
- A sequential color scale uses **light colors for low values** and **dark colors for high values**.
- It helps users quickly identify high and low marks.

4. Color Palette Suitable for Color-Blind Users

A good color-blind-friendly palette is **Viridis**.

Replace the color scale with:

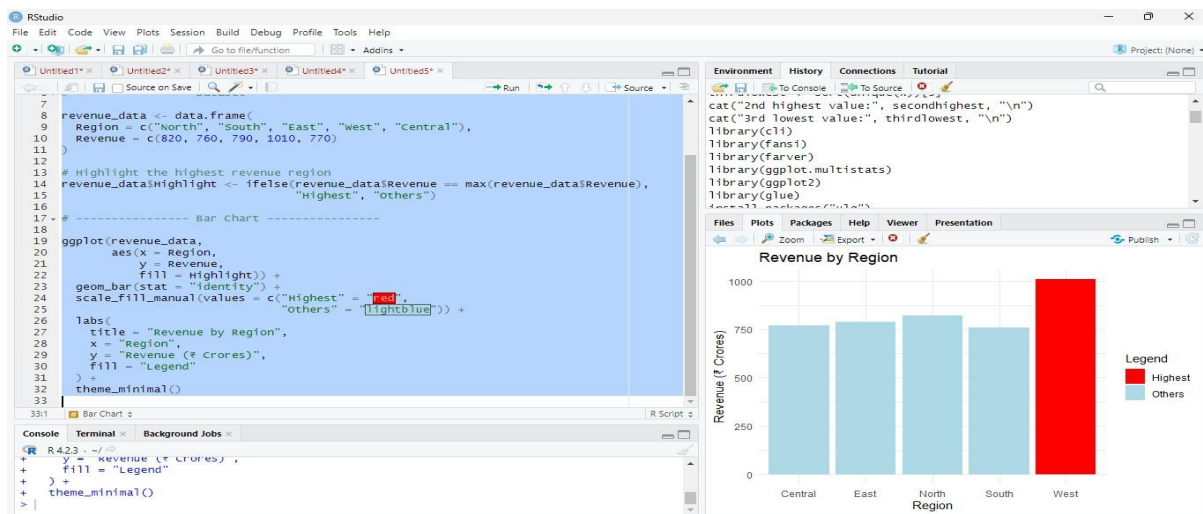
```
scale_fill_viridis_c()
```

(If needed, install the package first: `install.packages("viridis")`.)

5. Difference Between Sequential and Categorical Color Scales

Sequential Color Scale	Categorical Color Scale
Used for continuous numeric data	Used for categorical data
Colors change gradually from light to dark	Each category gets a distinct color
Example: Marks, Salary, Temperature	Example: Gender, Department, Product Category
Shows magnitude or intensity	Distinguishes between different groups

Q4: Highlighting with Color



2. How Highlighting Improves Data Interpretation

- Highlighting draws attention to the most important data point.
- It helps users quickly identify key insights.
- It makes the chart easier to understand at a glance.

3. Highest-Performing Region

- **West** has the highest revenue with **₹1010 Crores**.

4. Effect of Excessive Use of Colors

- Too many colors can make the chart confusing.
- Important information may become harder to identify.
- It reduces readability and distracts from the main message.

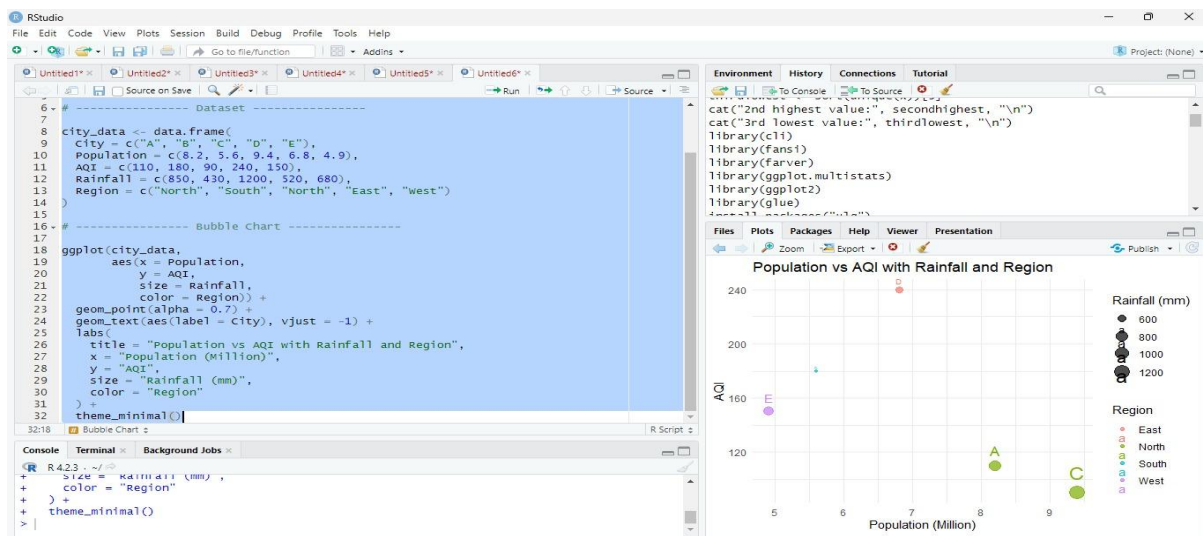
5. Alternative Highlighting Technique

Instead of color, you can:

- Add **data labels** (show the revenue value on each bar).
- Increase the **size** or **border thickness** of the highlighted bar.
- Use an **arrow** or **annotation** to point to the highest value.

These techniques emphasize important data without relying only on color.

Q5: Integrated Visualization



2. Aesthetic Mappings

Variable Aesthetic

Population X-axis

AQI Y-axis

Rainfall Bubble Size

Region Bubble Color

3. Interpretation of Population and AQI

- There is **no strong positive relationship** between Population and AQI.
- Some cities with larger populations have lower AQI, while smaller cities may have higher AQI.
- This suggests that factors other than population also affect air quality.

4. Continuous and Categorical Variables

Variable Type

Population Continuous

AQI Continuous

Rainfall Continuous

Region Categorical

City Categorical

5. Recommendation to Improve Readability

- Add **city labels** to each bubble (already included in the code using `geom_text()`).
- Increase bubble transparency (`alpha`) to reduce overlap.
- Use a **color-blind-friendly palette** (such as `scale_color_viridis_d()`) for better accessibility.